

Time Series Exercise Sheet 2

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Exercise 2.1

When we defined the MA(q) process we specified $\theta_0 = -1$. In terms of the covariance structure for moving average processes, why do we not gain more generality by letting θ_0 be an arbitrary number?

Solution 2.1

If we change the first coefficient, then we can divide through the specifying equation by that number and obtain a new process with the same autocorrelation. So for identifiability purpose, we force the coefficient to be $\theta_0 \equiv -1$.

For example, let's consider the first order moving average process, MA(1), when $\mu = 0$. In other words, the process is determined by $\theta_0 = -1$ and θ_1 :

$$X_t = -\theta_0\epsilon_t - \theta_1\epsilon_{t-1} = \epsilon_t - \theta_1\epsilon_{t-1}. \quad (1)$$

where $\epsilon_t \sim N(0, \sigma_\epsilon^2)$. Assuming there is another parametrization for the process using $\theta'_0 = c\theta_0 = c$ and θ'_1 for some $c \neq 0$.

$$X_t = -\theta'_0\epsilon'_t - \theta'_1\epsilon'_{t-1} = c\epsilon'_t - \theta'_1\epsilon'_{t-1}. \quad (2)$$

where $\epsilon'_t \sim N(0, \sigma_{\epsilon'}^2)$. If we rewrite (1) as

$$\begin{aligned} X_t &= \epsilon_t - \theta_1\epsilon_{t-1} \\ &= c\frac{\epsilon_t}{c} - (c\theta_1)\frac{\epsilon_{t-1}}{c}. \end{aligned}$$

Hence, $\theta'_1 = c\theta_1$ and $\epsilon'_t = c^{-1}\epsilon_t$ for all t . Moreover, $\sigma_{\epsilon'}^2 = c^{-2}\sigma_\epsilon^2$. Note that the two representations have the same autocovariance (and autocorrelation):

$$\text{Var}(X_t) = (1 + \theta_1^2)\sigma_\epsilon^2 \quad \text{from (1)}$$

$$\begin{aligned} \text{Var}(X_t) &= \{(\theta'_0)^2 + (\theta'_1)^2\}\sigma_{\epsilon'}^2 \\ &= c^2(1 + \theta_1^2)c^{-2}\sigma_\epsilon^2 = (1 + \theta_1^2)\sigma_\epsilon^2 \quad \text{from (2)} \end{aligned}$$

$$\text{Cov}(X_t, X_{t-1}) = \theta_0\theta_1\sigma_\epsilon^2$$

This shows us that there is an identifiability issue if we do not constrain $\theta_0 = -1$. Specifically, there are multiple MA(1) representations for X_t , making the representation of X_t non-unique.

Using the same argument to the general MA(q) process, we will reach the same conclusion as the MA(1) example.

Exercise 2.2

Determine that the moving average process defined by

$$X_t = \varepsilon_t - \theta\varepsilon_{t-1}$$

can be written as

$$X_t = \varepsilon_t - \sum_{j=1}^p \theta^j X_{t-j} - \theta^{p+1} \varepsilon_{t-p-1}$$

for any positive integer value of p .

Solution 2.2

Note that for any integer value of $p \geq 1$, we have $X_{t-p} = \varepsilon_{t-p} - \theta \varepsilon_{t-p-1}$, so

$$\varepsilon_{t-p} = X_{t-p} + \theta \varepsilon_{t-p-1}$$

Applying the latter recursively we have

$$\begin{aligned} X_t &= \varepsilon_t - \theta \varepsilon_{t-1} = \varepsilon_t - \theta (X_{t-1} + \theta \varepsilon_{t-2}) = \varepsilon_t - \theta X_{t-1} - \theta^2 \varepsilon_{t-2} \\ &= \varepsilon_t - \theta X_{t-1} - \theta^2 (X_{t-2} + \theta \varepsilon_{t-3}) \\ &= \varepsilon_t - \theta X_{t-1} - \theta^2 X_{t-2} - \theta^3 \varepsilon_{t-3} \\ &= \dots \\ &= \varepsilon_t - \sum_{j=1}^p \theta^j X_{t-j} - \theta^{p+1} \varepsilon_{t-p-1} \end{aligned}$$

Exercise 2.3

Assume that $Y_t = \sum_{s=-p}^p g_{s-t} \epsilon_s$ and $Z_t = \sum_{s=-p}^p h_{s-t} \epsilon_s$ where ϵ_t is zero-mean white noise. We define X_t pointwise by $Y_t + Z_t$. Determine the first and second moments of X_t .

Solution 2.3

Let $\text{Var}(\epsilon_s) = \sigma^2$ and rewrite $X_t = \sum_{s=-p}^p (g_{s-t} + h_{s-t}) \epsilon_s$. As ϵ_s are uncorrelated, we have for all $t, \tau \in T$

$$\begin{aligned} \mathbb{E}[X_t] &= \sum_{s=-p}^p (g_{s-t} + h_{s-t}) \mathbb{E}(\epsilon_s) = \sum_{s=-p}^p (g_{s-t} + h_{s-t}) \cdot 0 = 0 \\ \mathbb{E}[X_t X_{t+\tau}] &= \text{Cov}(X_t, X_{t+\tau}) = \text{Cov} \left(\sum_{s=-p}^p (g_{s-t} + h_{s-t}) \epsilon_s, \sum_{l=-p}^p (g_{l-t-\tau} + h_{l-t-\tau}) \epsilon_l \right) \\ &= \sum_{s=-p}^p \sum_{l=-p}^p (g_{s-t} + h_{s-t}) (g_{l-t-\tau} + h_{l-t-\tau}) \text{Cov}(\epsilon_s, \epsilon_l) \\ &= \sigma^2 \sum_{s=-p}^p (g_{s-t} + h_{s-t}) (g_{s-t-\tau} + h_{s-t-\tau}) \end{aligned}$$

Exercise 2.4

In the lectures we defined two different estimators for the ACVS:

$$\begin{aligned} \tilde{\gamma}_\tau &= \frac{1}{n - |\tau|} \sum_{t=1}^{n-|\tau|} \{X_t - \bar{X}\} \{X_{t+|\tau|} - \bar{X}\}, \\ \hat{\gamma}_\tau &= \frac{1}{n} \sum_{t=1}^{n-|\tau|} \{X_t - \bar{X}\} \{X_{t+|\tau|} - \bar{X}\}. \end{aligned}$$

If X_t has independent Gaussian realizations at each time point please calculate the mean, variance and mean-square error of these estimators. Repeat the calculation for an MA(1) process. As in class to do the analysis replace $\bar{X}$ by $\mu = \mathbb{E}(X_t)$, arguing that for large samples this will be appropriate.

(Hint): There is a known result (Isserlis' theorem) that if $\mathbf{X} = (X_1, X_2, X_3, X_4)$ is a multivariate normal random vector, then:

$$\mathbb{E}[X_1 X_2 X_3 X_4] = \mathbb{E}[X_1 X_2] \mathbb{E}[X_3 X_4] + \mathbb{E}[X_1 X_3] \mathbb{E}[X_2 X_4] + \mathbb{E}[X_1 X_4] \mathbb{E}[X_2 X_3].$$

Solution 2.4

For a white noise process we have (informally replacing $\bar{X}$ with μ like in class)

$$\begin{aligned}\mathbb{E}[\hat{\gamma}_\tau] &= \frac{1}{n} \sum_{t=1}^{n-|\tau|} \mathbb{E}[(X_t - \mu)(X_{t+|\tau|} - \mu)] \\ &= \begin{cases} \sigma_X^2 & \text{if } \tau = 0 \\ 0 & \text{otherwise.} \end{cases} \mathbb{E}[\tilde{\gamma}_\tau] = \begin{cases} \sigma_X^2 & \text{if } \tau = 0 \\ 0 & \text{otherwise.} \end{cases}\end{aligned}$$

Furthermore,

$$\begin{aligned}\text{Var}(\hat{\gamma}_\tau) &= \frac{1}{n^2} \sum_{t_1=1}^{n-|\tau|} \sum_{t_2=1}^{n-|\tau|} \text{Cov}((X_{t_1} - \mu)(X_{t_1+|\tau|} - \mu), (X_{t_2} - \mu)(X_{t_2+|\tau|} - \mu)) \\ &= \frac{1}{n^2} \sum_{t_1=1}^{n-|\tau|} \sum_{t_2=1}^{n-|\tau|} \{ \mathbb{E}[(X_{t_1} - \mu)(X_{t_1+|\tau|} - \mu)(X_{t_2} - \mu)(X_{t_2+|\tau|} - \mu)] \\ &\quad - \mathbb{E}[(X_{t_1} - \mu)(X_{t_1+|\tau|} - \mu)] \mathbb{E}[(X_{t_2} - \mu)(X_{t_2+|\tau|} - \mu)] \}\end{aligned}$$

Using Isserlis' theorem, we find

$$\begin{aligned}\mathbb{E}[(X_{t_1} - \mu)(X_{t_1+|\tau|} - \mu)(X_{t_2} - \mu)(X_{t_2+|\tau|} - \mu)] \\ = \gamma_X(|\tau|)\gamma_X(|\tau|) + \gamma_X(t_1 - t_2)\gamma_X(t_1 - t_2) + \gamma_X(t_1 - t_2 - |\tau|)\gamma_X(t_1 - t_2 + |\tau|)\end{aligned}$$

For white noise we have $\gamma_X(\tau) = \sigma_X^2 \delta_\tau$ where

$$\delta_\tau = \begin{cases} 1 & \tau = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore

$$\mathbb{E}[(X_{t_1} - \mu)(X_{t_1+|\tau|} - \mu)(X_{t_2} - \mu)(X_{t_2+|\tau|} - \mu)] = \sigma_X^4 \delta_\tau + \sigma_X^4 \delta_{t_1-t_2} + \sigma_X^4 \delta_{t_1-t_2} \delta_\tau \quad (3)$$

Thus we get

$$\text{Cov}((X_{t_1} - \mu)(X_{t_1+|\tau|} - \mu), (X_{t_2} - \mu)(X_{t_2+|\tau|} - \mu)) = \sigma_X^4 \delta_{t_1-t_2} + \sigma_X^4 \delta_{t_1-t_2} \delta_\tau.$$

We therefore find that

$$\text{Var}(\hat{\gamma}_\tau) = \frac{1 + \delta_\tau}{n^2} \sigma_X^4 (n - |\tau|).$$

It follows directly that

$$\text{Var}(\tilde{\gamma}_\tau) = \frac{1 + \delta_\tau}{n - |\tau|} \sigma_X^4.$$

The mean square error is then (for white noise)

$$\text{MSE}(\hat{\gamma}_\tau) = (0)^2 + \frac{n - |\tau|}{n^2} (1 + \delta_\tau) \sigma_X^4,$$

whilst

$$\text{MSE}(\tilde{\gamma}_\tau) = (0)^2 + \frac{1}{n - |\tau|} (1 + \delta_\tau) \sigma_X^4.$$

Thus in this case the biased estimator $\hat{\gamma}$ wins out.

For the MA(1) process, $\text{Cov}(X_t, X_{t+|\tau|}) = \delta_\tau (1 + \theta^2) \sigma_X^2 - \delta_{|\tau|-1} \theta \sigma_X^2$ and this goes into the calculations.

Exercise 2.5

Assume we know the mean of X_t and it is μ . Form the estimator

$$\bar{\gamma}_\tau^{(\alpha)} = \alpha \tilde{\gamma}_\tau + (1 - \alpha) \hat{\gamma}_\tau$$

For X_t Gaussian white noise determine the MSE of $\bar{\gamma}_\tau^{(\alpha)}$. For $\sigma_X^2 = 1$, $\tau = 1$ and $n = 10$ plot it as a function of α . What value of α is appropriate?

Solution 2.5

Wlog we may assume that $\mu = 0$, else we may define a new process, say $X_t^* = X_t - \mu$, s.t. $\gamma_{X^*} = \gamma_X$. For a white noise process, we note that both $\tilde{\gamma}$ and $\hat{\gamma}$ are unbiased, and hence $\bar{\gamma}^{(u)}$ is also unbiased, i.e. $\mathbb{E}[\bar{\gamma}_\tau^{(\alpha)}] = \gamma(\tau)$; see Exercise 2.4. Thus, for a white noise process, we may re-write the $\text{MSE}(\bar{\gamma}_\tau^{(\alpha)})$ as:

$$\begin{aligned} \text{MSE}(\bar{\gamma}_\tau^{(\alpha)}) &= \mathbb{E}[(\bar{\gamma}_\tau^{(\alpha)} - \gamma(\tau))^2] = \mathbb{E}[(\bar{\gamma}_\tau^{(\alpha)} - \mathbb{E}[\bar{\gamma}_\tau^{(\alpha)}])^2] = \text{Cov}(\bar{\gamma}_\tau^{(\alpha)}, \bar{\gamma}_\tau^{(\alpha)}) \\ &= \text{Cov}(\alpha \tilde{\gamma}_\tau + (1 - \alpha) \hat{\gamma}_\tau, \alpha \tilde{\gamma}_\tau + (1 - \alpha) \hat{\gamma}_\tau). \end{aligned}$$

We continue by expanding the terms inside the covariance

$$\text{MSE}(\bar{\gamma}_\tau^{(\alpha)}) = \alpha^2 \text{Cov}(\tilde{\gamma}_\tau, \tilde{\gamma}_\tau) + 2\alpha(1 - \alpha) \text{Cov}(\tilde{\gamma}_\tau, \hat{\gamma}_\tau) + (1 - \alpha)^2 \text{Cov}(\hat{\gamma}_\tau, \hat{\gamma}_\tau). \quad (4)$$

We use the fact that $\hat{\gamma}_\tau = \frac{n-|\tau|}{n} \tilde{\gamma}_\tau$ to compute

$$\begin{aligned} \text{Cov}(\tilde{\gamma}_\tau, \hat{\gamma}_\tau) &= \frac{n-|\tau|}{n} \text{Var}(\tilde{\gamma}_\tau), \\ \text{Var}(\hat{\gamma}_\tau) &= \left\{ \frac{n-|\tau|}{n} \right\}^2 \text{Var}(\tilde{\gamma}_\tau). \end{aligned}$$

For the time being, assume that $\text{Var}(\tilde{\gamma}_\tau) = V(\tau)$, and let $C(\tau) = \frac{n-|\tau|}{n}$. Then we note

$$\text{MSE}(\bar{\gamma}_\tau^{(\alpha)}) = (\alpha - \alpha C(\tau) + C(\tau))^2 V(\tau).$$

A good choice of α can be computed by minimizing $\text{MSE}(\bar{\gamma}_\tau^{(\alpha)})$. Since $V(\tau)$ is not dependent on α for a given τ , we simply minimize $(\alpha - \alpha C(\tau) + C(\tau))^2$. We have for $\tau > 0$ that $C(\tau) < 1$ hence the expression in brackets is always positive for $|\tau| < n$. Differentiation wrt. α yields

$$(\alpha - \alpha C(\tau) + C(\tau))(1 - C(\tau)) \stackrel{!}{=} 0,$$

from where, for $\tau > 0$, we can pick $\alpha = C(\tau)/(C(\tau) - 1)$. For $\tau = 0$ any choice of α leaves the MSE unchanged.

We demonstrate this for the case when X_t is a Gaussian white noise. From Exercise 2.4, we already know $\text{Var}(\tilde{\gamma}_\tau) = \frac{1+\delta_\tau}{n-|\tau|} \sigma_X^4$. Finally, for $\tau > 0$, replacing $n = 10$, $\sigma_X = 1$ we obtain

$$\text{MSE}(\bar{\gamma}_\tau^{(\alpha)}) = \left\{ \alpha - \alpha \frac{n-|\tau|}{n} + \frac{n-|\tau|}{n} \right\}^2 \frac{1}{n-|\tau|} \quad (5)$$

In Figure 1 we plot the MSE for $\tau = 1$ as a function of α .

Exercise 2.6

Show that for any series $\{x_1, \dots, x_n\}$ the sample autocovariance satisfies $\sum_{|\tau| < n} \hat{\gamma}_\tau = 0$.

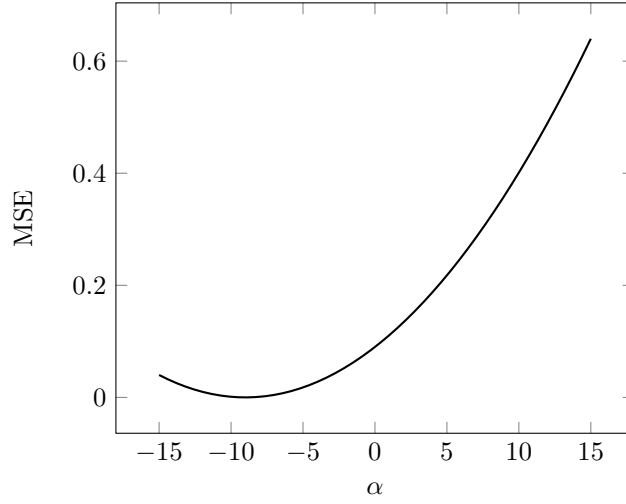


Figure 1: MSE of the autocovariance estimate for different values of α .

Solution 2.6

Let $\bar{X} = (\sum_{t=1}^n X_t)/n$. We compute:

$$\begin{aligned}
 \sum_{\tau=-n+1}^{n-1} \hat{\gamma}_{\tau} &= \frac{1}{n} \sum_{\tau=-n+1}^{n-1} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}) \\
 &= \frac{1}{n} \sum_{t_1=1}^n \sum_{t_2=1}^n (X_{t_1} - \bar{X})(X_{t_2} - \bar{X}) \\
 &= \frac{1}{n} \left(\sum_{t=1}^n (X_t - \bar{X}) \right)^2 \\
 &= 0.
 \end{aligned}$$

Exercise 2.7

Prove that, for a causal AR(p), we have

$$\alpha_{\tau} = 0, \quad \forall \tau > p.$$

Hint: use the prediction equations which say that best linear prediction satisfies

$$\text{Cov}(Y - \mathcal{P}_{X_1, \dots, X_n}(Y), X_k) = 0 \quad (6)$$

for all $k \in \{1, \dots, n\}$. Also note that for causal processes $\text{Cov}(X_t, \epsilon_s) = 0$ when $t < s$.

Solution 2.7

Consider $\tau > p$. We know from the lectures that $\alpha_{\tau} = \alpha_{\tau, \tau}$ in

$$\mathcal{P}_{X_0, \dots, X_{\tau-1}}(X_{\tau}) = \sum_{j=1}^{\tau} \alpha_{\tau, j} X_{\tau-j}. \quad (7)$$

All that remains is to show that this is zero. In particular, we know from the prediction equations that

$$\text{Cov}(X_{\tau} - \mathcal{P}_{X_0, \dots, X_{\tau-1}}(X_{\tau}), X_k) = 0, \quad \forall k \in \{0, \dots, \tau-1\}. \quad (8)$$

Furthermore, we have $\forall k \in \{0, \dots, \tau - 1\}$

$$\text{Cov} (X_\tau - \mathcal{P}_{X_0, \dots, X_{\tau-1}}(X_\tau), X_k) = \text{Cov} \left(X_\tau - \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}, X_k \right) \quad (9)$$

$$= \text{Cov} \left(\sum_{j=1}^p \phi_j X_{\tau-j} + \varepsilon_t - \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}, X_k \right) \quad (10)$$

$$= \sum_{j=1}^{\tau} (\tilde{\phi}_j - \alpha_{\tau,j}) \text{Cov} (X_{\tau-j}, X_k) + \text{Cov} (\varepsilon_t, X_k) \quad (11)$$

where $\tilde{\phi}_j = \phi_j$ for $j \leq p$ and $\tilde{\phi}_j = 0$ for $j > p$.

Now $\text{Cov} (\varepsilon_t, X_k) = 0$, so if $(\tilde{\phi}_j - \alpha_{\tau,j}) = 0$ for all j , then the covariance is zero and we have the best linear predictor.

Exercise 2.8

For the 6 time series shown in Figure 2, identify if any of the following models could be appropriate

1. White noise,
2. MA(q),
3. AR(p),
4. ARMA(p, q).

Give your reasoning.

Solution 2.8

The true underlying processes are as follows:

1. ARMA(1, 1),
2. MA(1),
3. AR(1),
4. white noise,
5. not stationary,
6. AR(1).

However, the important point is that you make a sensible argument for your choice in each case. Example arguments for some reasonable models are given below:

1. Looking at the time series, it looks like the mean and variance are not changing over time. Furthermore, both the ACF and PACF decay in magnitude and do not sharply drop off after some point. Therefore, we might suggest an ARMA model.
2. Again the time series seems to be stationary. We see a sharp drop in the ACF after lag 1, and a decay in the PACF. This suggests an MA(1) model is appropriate.
3. The process looks to be stationary, the ACF decays in magnitude as lag increases, but the PACF drops after lag 1. Therefore an AR(1) model is likely appropriate.
4. The process seems stationary, and both the PACF and ACF are not significant for non-zero lags. Therefore a white noise model is likely to be appropriate.
5. This process has very strong lag 1 PACF, and the ACF decays very slowly. In conjunction, the time series looks as though it could have a changing variance, so this is likely not to be stationary. This should be explored further, but a stationary model is likely not appropriate.

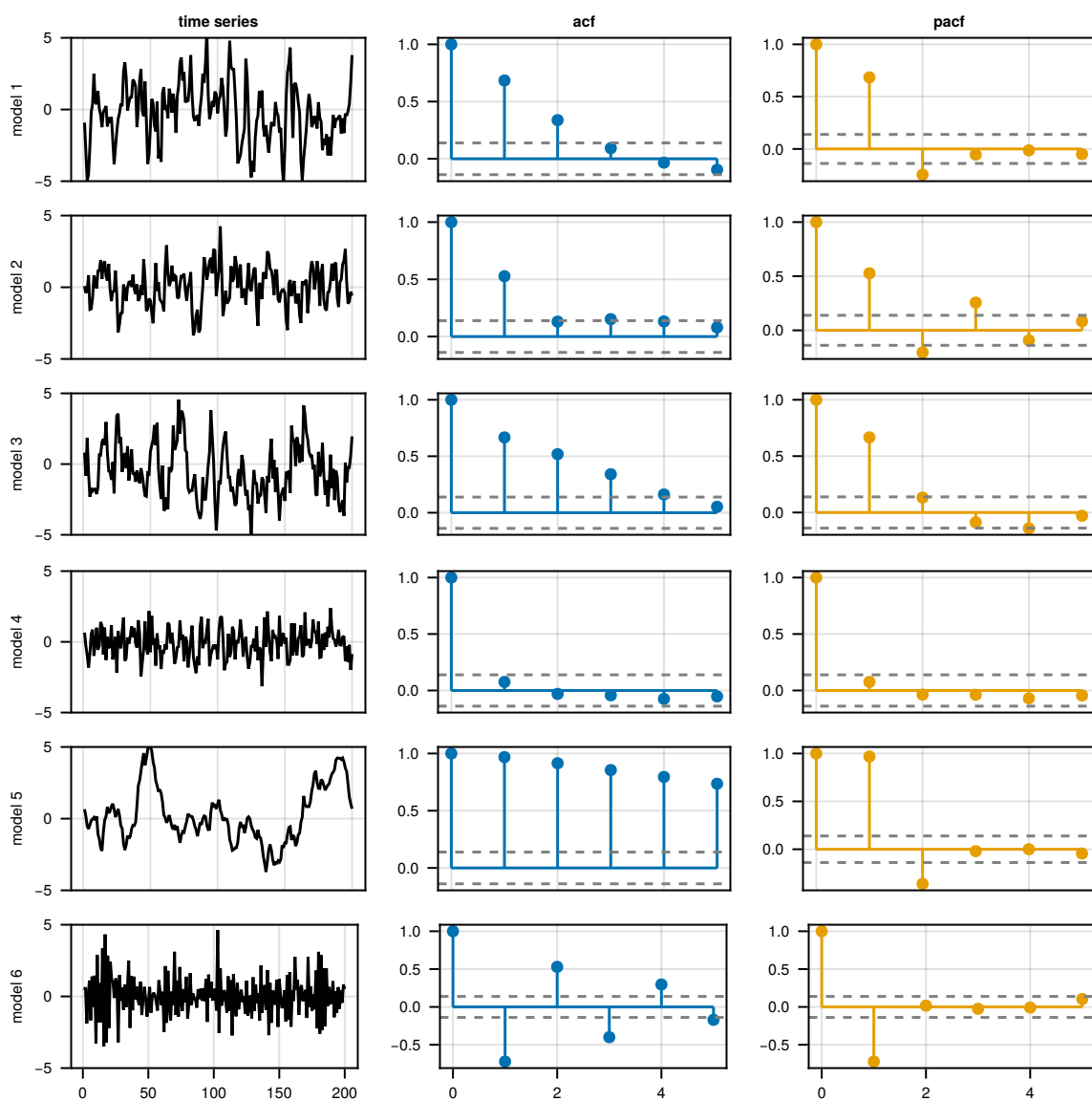


Figure 2: Time series and their sample acf and pacfs

6. This process looks stationary, the ACF decays in magnitude, but the PACF drops to non-significant after lag 1. Therefore an AR(1) model may be appropriate.

Note that in some of these cases, one could argue for a different model. For example, one could possibly argue for an AR(2) in the first example, or possibly an ARMA model in the third example. One could also argue for an AR(1) with very high correlation in the fifth example. The purpose of this exercise is to practice making such arguments, and to realise that model selection is not trivial! There are other forms of model selection one can use to choose an appropriate model, some of which will be seen later in the course. But looking at the time series, the ACF and the PACF are useful sanity checks in any case.